

CLAIRE: a Controllable LLM Tutoring Framework for Reading Comprehension

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Abstract. Reading comprehension requires strategic and metacognitive support provided by guided reading methodologies through structured, scaffolded interaction. However, enacting these methodologies systematically at scale remains challenging, particularly in digital contexts. Recent advances in Large Language Models (LLMs) open new opportunities, though existing approaches largely rely on unstructured dialogue and implicit pedagogical control. In this paper, we present CLAIRE, a teacher-configurable framework that operationalizes guided reading by decomposing the pedagogical procedure into four components, i.e., phases, interaction moves, assessment criteria, and transition logic, authored by teachers and executed at runtime through a structured multi-agent dialogue controller. Experimental evaluation shows that CLAIRE can faithfully enact a teacher-defined guided reading methodology and is perceived by teachers as pedagogically meaningful and useful, supporting the scalable adoption of structured reading instruction with LLMs.

Keywords: Intelligent Tutoring Systems · Large Language Models · Multi-Agent Systems · Reading Comprehension.

1 Introduction

Reading comprehension is widely understood as an active process in which readers construct and integrate meaning through inference, monitoring, and strategic regulation [6,15]. However, many of the strategies required for deep comprehension, such as prediction, clarification, and summarization, are not consistently activated by young readers and require explicit instructional support. As a result, comprehension difficulties often reflect limited support for strategic and metacognitive processes rather than lack of access to textual information [3].

Instructional approaches such as guided reading have long sought to address this gap by structuring reading activities into phases and by scaffolding strategy use through targeted interaction. In practice, however, these approaches are difficult to enact systematically. Effective scaffolding requires continuous adaptation to student responses and fine-grained decisions about when to persist, rephrase, or advance, that are hard to sustain at scale and are often left implicit in classroom practice. These challenges are further amplified in digital reading contexts, which tend to promote faster, less reflective reading and weaker

monitoring [30,22]. Consequently, pedagogical procedures remain inconsistently applied and difficult to customize to individual learners.

Recent advances in Large Language Models (LLMs) offer new opportunities to address these limitations by supporting the procedural execution of guided reading methodologies. Although recent work has explored LLM-based and agentic systems for reading comprehension and question answering [11,7,27], existing approaches lack structured, multi-phase control. To effectively support reading instruction, however, AI-based tutoring systems should move beyond the role of general-purpose dialogue partners and instead operationalize reading as an explicit instructional process. This entails: (i) structuring reading into clearly defined phases; (ii) instantiating different types of interaction aligned with phase objectives; (iii) dynamically selecting interaction strategies based on learner responses; (iv) supporting automated assessment of student contributions; and (v) enabling customization of the instructional process to individual learners. Crucially, the role of AI in this context is not to replace comprehension instruction, but to make it systematic and inspectable by explicitly enacting its underlying pedagogical structure.

In this paper, we introduce CLAIRE¹, Conversational Literacy Agent Integrating Reflective Engagement, a teacher-configurable framework for guided reading comprehension that treats instructional dialogue as a controllable process rather than as a monolithic generative interaction. CLAIRE operationalizes guided reading by decomposing the pedagogical procedure into separable components that specify (i) what the learner is expected to do at each moment (phases), (ii) how the tutor should respond (interaction moves), (iii) how student contributions are evaluated (assessment criteria), and (iv) when the system should persist, change strategy, or advance to the next phase (transition logic).

Rather than encoding instructional decisions implicitly in a single prompt, CLAIRE represents them explicitly and executes them at runtime through a structured multi-agent dialogue controller. This design makes the instructional flow systematic and inspectable, while allowing teachers to define phases, interaction moves, assessment criteria, and transition rules to instantiate different guided reading methodologies without modifying system code.

To assess the effectiveness of our approach, we conducted an experimental evaluation that investigates both the learning experience and the quality of the automated tutoring behavior. In particular, we examine teachers’ perceptions of interaction quality, support for reading comprehension, and pedagogical usefulness, as well as the system’s ability to correctly enact a teacher-defined guided reading comprehension methodology. The experimental results indicate that the proposed framework can effectively support guided reading comprehension activities by making pedagogical procedures explicit, configurable, and inspectable, while leveraging LLMs to enable flexible and meaningful interactions. Moreover, the framework can facilitate the scalable adoption of structured instructional methodologies.

¹ <https://anonymous.4open.science/r/Claire-82D2>

2 Related Work

Teaching Reading Comprehension. Reading comprehension cannot be reduced to memorizing information or reproducing the surface structure of a text. Rather, it involves actively constructing meaning by integrating textual information with prior knowledge [15,8]. According to established cognitive models, comprehension entails the progressive construction of coherence by identifying semantic relations and making necessary inferences [15,16]. Research on skilled readers has shown that these processes are supported by strategic behaviors such as making predictions, generating questions, clarifying misunderstandings, and synthesizing key information, which render cognitive processes explicit and regulable during reading [26]. Consequently, effective instruction in reading comprehension emphasizes the explicit teaching of such strategies, not as rigid procedures but as flexible cognitive tools that support active engagement and self-regulation in meaning construction. This instructional need becomes particularly salient in digital reading contexts, where faster, more fragmented reading often reduces metacognitive monitoring [1,30,22].

Automatic approaches to Reading Comprehension. Computer-assisted approaches to reading comprehension have long demonstrated that human-computer interaction can enhance the pedagogical value of reading activities [2]. Early work focused on web-based interactive systems without artificial intelligence, offering automatic feedback for reading aloud [25], multiple representations to support deaf students [24], and scaffolding for expository and inferential skills [36]. Additional systems identified signal words and main ideas to strengthen problem-solving abilities [13] and provided explicit strategy instruction [28,32]. Interaction in these environments was largely scripted rather than adaptive, relying on rule-based feedback mechanisms such as detecting hesitation and prompting the next words during oral reading [25], automatically assessing vocabulary and grammatical accuracy [36], or delivering formative feedback during activities and on answer correctness [28].

More recent work has incorporated artificial intelligence to support reading comprehension. Early AI-based systems leveraged natural language processing to model and predict student knowledge [10], automatically assess student performance [23,35], or generate templated questions using techniques such as coreference resolution and dependency parsing [18]. Subsequent research explored chatbot-based approaches, either by training task-specific generative models [37,33,9] or by adopting zero-shot generative models. The latter has become the dominant paradigm, enabling personalized feedback, automatic generation of skill-based questions [11,34], text simplification [4], supplementary learning materials [17], automatically generated activities [12], student profiling and exercise recommendation [20], and Socratic-style questioning [7].

However, most existing approaches rely on open-ended interactions with limited control over the activation of strategic and metacognitive processes critical for deep reading comprehension [3,26]. Even recent LLM-based and agentic systems [5,27,31,14] are primarily applied to learning-by-teaching or treat gener-

ative models as general-purpose dialogue agents, without explicitly structuring instructional scaffolding as a phase-based, teacher-configurable process that integrates assessment and systematically orchestrates tutoring strategies.

In contrast, CLAIRE is explicitly designed for reading comprehension, operationalizing instructional scaffolding and making the tutoring process inspectable and controllable, so teachers can verify how feedback and strategy use relate to instructional objectives.

3 Approach

CLAIRE is designed to support structured instructional practices grounded in established pedagogical frameworks for reading comprehension. Its primary goal is to enable teachers to operationalize effective teaching strategies into explicit, controllable, and repeatable interaction patterns. To this end, instruction is organized into steps, each corresponding to a well-defined phase of the learning process, rather than being embedded implicitly in a generic dialogue model. Each phase defines a learning objective and is enacted through one or more cycles of interaction and evaluation designed to progressively refine the learner’s understanding. This iterative interaction-assessment loop enables the system to support learning as a gradual process rather than a single-turn outcome. Depending on the evaluated adequacy of the student’s response, the system may persist within the same phase, adapt the instructional strategy by selecting a different interaction move, or advance to the next phase once the phase objective is met. By organizing instruction around bounded cycles of elicitation, feedback, and strategy adaptation, CLAIRE mirrors the stepwise procedures adopted in structured pedagogical frameworks and supports targeted scaffolding aimed at improving student comprehension over successive interactions.

3.1 Overview of the workflow

As illustrated in Figure 1, the tutoring process is implemented as a modular controller composed of four cooperating agents, each responsible for a distinct pedagogical function.

Learning activities are organized into teacher-defined phases, which constitute the fundamental unit of instruction. Each phase is operationalized by the **Phase Agent**, responsible for instructional framing. At the beginning of each phase, CLAIRE feeds to the agent the pedagogical objective, the student-facing instructions and the story text. This framing step establishes the instructional context and makes the expected learning outcome explicit. Once a phase is active, the system engages the learner in a bounded sequence of student-tutor interactions, regulated by a predefined interaction budget n_p that limits the number of turns allowed within each phase. Tutor utterances are generated by the **Interaction Agent**, which executes a teacher-defined interaction move. Each move operationalizes a specific pedagogical strategy, such as *mirroring*, *stimulation*, or *thinking aloud*, and constrains how the tutor should respond to

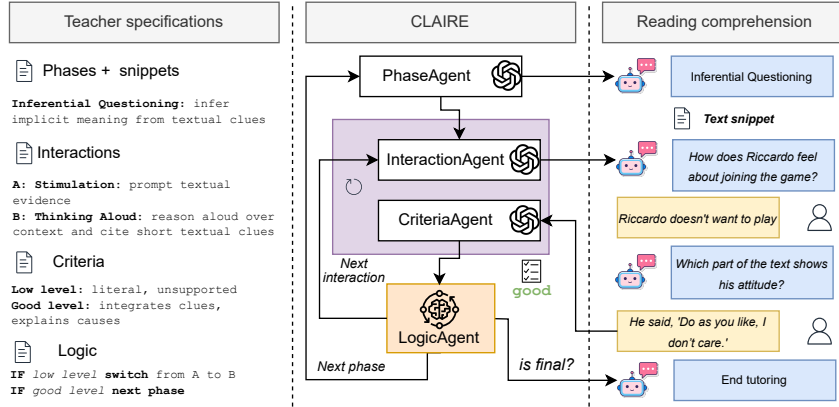


Fig. 1: Instantiation of the CLAIRE multi-agent tutoring workflow in a reading-comprehension scenario, structured around teacher specifications, underlying agent components, and the conversational interface.

the student’s contribution. After each student response, the **Criteria Agent** evaluates the contribution using a rubric defined by the teacher for the current phase. Instead of generating free-form feedback, the agent assigns a discrete evaluation level that captures the adequacy of the response with respect to the phase objective, enabling consistent assessment and providing a structured signal for subsequent control decisions. Moreover, an additional *not adequate* rubric level is hard-coded into the Criteria Agent, relieving teachers from the need to explicitly define labels to detect off-topic or irrelevant student contributions. The **Logic Agent** governs the progression of the activity by deciding, based on the current phase, the selected interaction move, the evaluation level produced by the Criteria Agent, and the number of attempts already made, whether to continue within the current phase with a new interaction move or to advance to the next phase. Inside each phase, CLAIRE is forced to advance to the next phase if the number of attempts reaches the teacher-defined interaction budget n_p . Each interaction classified as *not adequate* by the Criteria Agent does not count toward the total number of attempts.

The **Phase**, **Interaction**, and **Criteria** Agents are implemented as LLM-based components, each operating under a role-specific prompt and receiving only the information needed for its function (e.g., phase objectives, move directives, or rubric definitions), which enables modular prompting and isolates the effects of individual components. In contrast, the **Logic Agent** is non-LLM and deterministically manages phase progression according to the teacher-defined sequence, ensuring reproducible dialogue flow and preventing instructional drift.

Table 1: Guided-reading configuration. For each phase, we report the available interaction types, the adequacy levels used for evaluation, and the maximum number of interaction attempts (n_p).

Repeat	Phase	Interaction types	Criteria (<i>adequacy</i> levels)	n_p
	Story Prediction	Mirroring	Not adequate; Superficial; Inferential	1
×2	Clarification	Thinking Aloud	Not adequate; Adequate	1
	Questioning	Mirroring	Not adequate; Superficial; Inferential	1
	Inferential Questioning	Mirroring; Stimulation; Thinking Aloud	Not adequate; Superficial; Inferential	3
	Summary	Mirroring; Stimulation; Thinking Aloud	Not adequate; Superficial; Inferential	3

3.2 Teacher specification

All pedagogical and control decisions in CLAIRE are configured by teachers through a set of spreadsheets, which together define the tutoring controller. The *phases spreadsheet* specifies each phase through five fields: a phase identifier, a pedagogical objective, a phase description, the associated snippet of the story text (displayed to the student and not editable by the system), and a maximum number of interaction attempts. The *interactions spreadsheet* defines the available interaction moves for each phase, specifying a move identifier, the phase to which it applies, and a directive that describes how the tutor should respond when the move is selected. The *criteria spreadsheet* encodes the assessment rubric for each phase, defining one or more evaluation criteria along with their discrete levels and level descriptions. Finally, the *logic spreadsheet* specifies the transition policy by mapping the current phase, the last interaction move, and the assessed evaluation level to either (i) a subsequent interaction move within the same phase or (ii) a transition to the next phase.

Spreadsheet entries are compiled into prompts and structured control variables that are passed among agents at runtime. This configuration cleanly separates pedagogical design from system implementation: teachers can modify phase structures, teaching strategies, assessment rubrics, and transition rules without changing the codebase, enabling rapid experimentation with different pedagogical frameworks and instructional designs.

4 Case study: *The Captain of the Courtyard*

We instantiate the system to enact a structured guided reading methodology for story-based reading comprehension, focusing on prediction, clarification, questioning, and summarization. The selected text, *The Captain of the Courtyard*, narrates the story of Riccardo, a boy who has recently joined a new class and is encouraged by his classmates to participate in a sports tournament. Through this experience, Riccardo gradually becomes part of the team and the class, conveying a message about inclusion, collaboration, and shared goals. Following the phase-based control model introduced in Section 3.1, the activity is instantiated through the **Phase Agent** as a sequence of eight phases, summarized in Table 1. The story text is divided into two parts, resulting in the repetition of

<i>Superficial Criterion</i>	<i>Mirroring Interaction</i>
A short, vague, or off-topic response that shows no personal elaboration and limits itself to the surface of the text. Classify the response as L1 – superficial when the student predominantly: (i) repeats words or phrases from the text without adding explanations or interpretations; [...] (v) provides literal information without showing connections, motivations, or deductions. Examples: "Because.", [...], "Maybe... I don't know..."	The chatbot receives the child's question and rephrases it briefly, faithfully, and non-judgmentally, demonstrating active listening and respect for their curiosity. [...] The tone is calm, welcoming, cooperative, and curious, typical of an educational dialogue. As an example, the message can be formulated as follows: "You're saying that [faithful rephrasing]. That could help us better review what we've read."

(a) Description of the *superficial* criterion (b) Description of the *mirroring* move

Fig. 2: Examples of teacher-defined assessment criterion (*superficial*) and interaction move (*mirroring*). These descriptions are provided to the LLM, together with the chat history, to guide student evaluation and tutoring behavior.

three phases (*Clarification*, *Questioning*, and *Inferential Questioning*) for each text segment. To regulate the interaction and ensure controlled progression, each phase is associated with a teacher-defined interaction budget n_p . In this instantiation, the budget is set to $n_p = 1$ for all phases except *Inferential Questioning* and *Summary*, which allow for multiple attempts ($n_p = 3$) to support deeper reasoning and synthesis. This configuration enables targeted scaffolding while keeping the overall interaction length bounded.

The activity begins with a *Story Prediction* phase, in which the student infers the content of the story from its title. It then proceeds through two repeated guided blocks, reflecting the iterative application of the same instructional structure. Each block includes a *Clarification* phase, aimed at resolving lexical or conceptual difficulties; a *Questioning* phase, where the student formulates a question about the text; and an *Inferential Questioning* phase, which supports core comprehension through scaffolded inferential prompts. The activity concludes with a *Summary* phase, in which the student produces a coherent retelling of the entire story and reflects on its main ideas.

As summarized in Table 1, the **Interaction Agent** executes three interaction move types: *Mirroring*, used to acknowledge and rephrase student contributions; *Stimulation*, used to prompt further elaboration; and *Thinking Aloud*, used to model text-grounded reasoning when student responses remain superficial. These interaction moves are selectively applied across phases according to the teacher-defined configuration.

Student contributions are evaluated by the **Criteria Agent** using a single assessment criterion, *Adequacy*, defined by the teacher. The criterion includes three rubric levels: (i) *not adequate* responses, which are unrelated to the task; (ii) *superficial* responses, which are vague, literal, or weakly supported; and (iii) *inferential* responses, which integrate textual clues and draw meaningful conclusions. In the *Clarification* phase, the rubric is reduced to two levels, with

superficial and *inferential* responses merged into a single *adequate* category. This design choice reflects the nature of the phase, in which students are not expected to produce inferential interpretations but rather to indicate which parts of the text are unclear.

The discrete evaluation produced by the Criteria Agent is consumed by the **Logic Agent**, which decides whether to continue scaffolding within the current phase or advance to the next one, subject to the teacher-defined interaction budget n_p . In this instantiation, advancement occurs when a student response is classified as *inferential*, while lower-level responses trigger the selection of a new interaction move. Figure 2 illustrates the teacher-defined assessment criteria and interaction moves used in this configuration. The complete set of specifications provided to the agents is available in the GitHub repository.

5 Experimental Analysis

The evaluation addresses two complementary research questions: Section 5.1 focuses on the learning process and user experience, capturing teachers’ perceptions of tutoring quality, support for reading comprehension, perceived adaptivity, pedagogical usefulness, and classroom acceptability. Section 5.2 evaluates the fidelity of the automated implementation, examining whether CLAIRE correctly enacts a teacher-defined guided reading comprehension methodology in the system instance described in Section 4, without assessing the pedagogical validity of the methodology itself. In the experiments, CLAIRE is powered by **gpt-4o-mini**, chosen to combine strong text understanding with low API costs and generation latency.

5.1 Process-related evaluation: learning experience and perception

The process-related evaluation examines how the interaction with CLAIRE is perceived in terms of learning experience and instructional value. We therefore conducted a focus-group-based evaluation involving 29 participants with pedagogical training (i.e., prospective and in-service teachers), who interacted with the system while assuming the role of students. This setup allowed participants to experience the tutoring process and to assess its alignment with instructional expectations and teaching practices. Participants’ perceptions were collected through a structured post-interaction questionnaire. Among the participants, 10.3% reported no prior teaching experience. Of those with teaching experience, 3.8% had less than 1 year of experience, 73.1% reported between 1 and 5 years, while 23.1% reported between 6 and 10 years. The majority (80.8%) had teaching experience in primary school. Table 2 reports the results of the evaluation, summarizing the questionnaire items and the mean ratings on a 5-point Likert scale ranging from 1 (very negative) to 5 (excellent).

The questionnaire captures participants’ perceptions along three macro-categories: **Tutoring Quality**, which concerns the quality of the interaction independently of its pedagogical impact; **Support for Reading Comprehension**,

Table 2: Teacher questionnaire results. For each construct (shown in bold), the corresponding questionnaire items are listed. Numerical values are reported at the construct level and represent mean ratings on a 5-point Likert scale.

		1	2	3	4	5
Tutor Quality	Clarity and Communicative Appropriateness	1.7	0.0	15.5	<u>32.8</u>	50.0
	The questions asked by CLAIRE are clear and easy to understand \ The agent’s tone is supportive and non-judgmental.					
	Response Relevance and Coherence	0.0	13.8	24.1	31.0	31.0
	The agent’s responses are relevant and coherent with what the student says.					
Reading Comprehension	Interaction Naturalness	0.0	0.0	37.9	<u>34.5</u>	27.6
	The interaction with CLAIRE feels natural and conversational rather than test-like.					
	Support for the Comprehension Process	0.0	8.6	19.0	39.7	<u>32.8</u>
	CLAIRE helps the student focus on the most important parts of the text \ The interaction encourages a more careful and reflective reading.					
Teachers’ Perceptions	Perceived Adaptivity to Student Responses	0.0	6.9	<u>34.5</u>	41.4	17.2
	I have the impression that CLAIRE adapts its interventions based on the student’s responses.					
	Perceived Pedagogical Usefulness	0.0	8.6	<u>29.3</u>	38.0	24.1
	The interaction with the agent makes aspects of the student’s comprehension process visible that would otherwise be difficult to observe \ The downloadable Excel report provides information that is useful for formative assessment.					
	Affective and Contextual Acceptance	2.5	22.8	21.8	29.9	<u>23.0</u>
	I would feel comfortable using CLAIRE during classroom activities \ CLAIRE could be integrated into the reading comprehension activities I usually carry out in class \ The agent’s style is appropriate for working with primary school students.					

which focuses on the perceived effects of the interaction on students’ comprehension processes; and **Teachers’ Perceptions**, which captures the system’s perceived pedagogical usefulness and contextual acceptability.

Tutoring Quality. We evaluated three constructs: *Clarity and Communicative Appropriateness*, *Response Relevance and Coherence*, and *Interaction Naturalness*. Overall, ratings are concentrated in the upper part of the scale, with values 4–5 accounting for the majority of responses across all three constructs. In particular, *Clarity and Communicative Appropriateness* received the most positive evaluations, with 82.8% of responses in the upper range of the scale (ratings 4–5). Similarly, *Interaction Naturalness* was positively perceived, with 62.1% of responses rated as 4–5. By contrast, *Response Relevance and Coherence* exhibited slightly lower performance, with 62.0% of responses in the 4–5 range, and a higher proportion of mid-scale ratings. We attribute this effect primarily to the current chat-history management strategy, which provides the full dialogue to the LLM introducing verbosity and less relevant context in longer interactions [19], as well as to occasional suboptimal decisions by the Criteria Agent when determining whether to deepen or redirect instructional focus (see Section 5.2). While larger language models could mitigate these issues [21], this would involve trade-offs in terms of generation latency and student engagement.

Table 3: Macro-averaged precision (P), recall (R), and F1 of the Criteria Agent by phase. † indicates phases with two rubric levels only.

Phase	Macro P	Macro R	Macro F1	Support
Prediction	77.45	77.45	77.45	23
Clarification [†]	100.00	100.00	100.00	69
Questioning	74.76	74.91	73.40	94
Inferential questioning	93.52	82.50	85.93	67
Summary	81.40	77.28	78.05	69
Average	85.43	82.43	82.97	-

Support for Reading Comprehension. This dimension evaluates the extent to which the questions generated by CLAIRE are perceived as supporting students’ reading comprehension processes, including attentive reading, reflection on the text, inference-making, and perceived adaptivity to student responses. Results indicate a positive assessment in this area. A large majority of respondents report that the interaction supports focused and reflective reading, with 72.5% of responses falling in the upper range of the scale (ratings 4–5) for items related to attentive reading. Moreover, the system is perceived as exhibiting a good degree of adaptivity to student responses (58.6% ratings of 4–5). While adaptivity is a key component of the tutoring experience, it likely reflects the combined effects of the underlying LLM and the teacher-defined prompting strategy.

Teachers’ Perceptions. Results indicate a generally positive perception of CLAIRE as a support for instructional practice. Participants report that the system helps make otherwise hard-to-observe aspects of students’ comprehension processes visible, with 62.1% of responses in the upper range (ratings 4–5) for items related to perceived pedagogical usefulness. The system is also viewed as largely acceptable in classroom contexts, with 52.9% of responses rated 4–5 for affective and contextual acceptance, and more than half of the participants reporting willingness to use it even in its current prototypical form. At the same time, some variability in acceptance emerged, suggesting that perceptions may depend on factors such as the agent’s interaction style, its compatibility with existing classroom routines, and individual differences in teachers’ familiarity and confidence with educational technologies.

5.2 Task-related evaluation: assessing the automation

The task-related evaluation assesses whether CLAIRE correctly automates a teacher-defined guided reading comprehension methodology. Rather than evaluating the pedagogical soundness of the method, it focuses on the fidelity and reliability of its LLM-based implementation. Three dimensions are considered: correctness of student assessment, alignment with the pedagogical method, and robustness to instructional drift.

Evaluation of the Criteria Agent. To assess the correctness of student evaluations produced by the Criteria Agent, system-assigned assessment levels were

compared with human annotations following the same criteria. Table 3 reports macro-averaged precision, recall, and F1 scores for each instructional phase. Performance is generally high, with macro F1 ranging from 73.4% for *Questioning* to 85.9% for *Inferential questioning*, and an overall average F1 of 83.0%. The highest consistency with human judgment is observed in *Clarification* interactions, where precision, recall, and F1 reach 100%, while *Questioning* shows slightly lower performance, likely reflecting the greater variability in responses. These results demonstrate that the Criteria Agent reliably reproduces teacher-like evaluations across multiple phases of the guided reading process.

Alignment with the pedagogical method. In CLAIRe, key aspects of the tutoring process are fixed by design. This evaluation therefore focuses on the alignment of runtime-generated interactions with the intended pedagogical method. To assess this alignment, teachers rated the appropriateness of responses produced by the Interaction Agent on a 5-point Likert scale, evaluating whether they supported higher-level comprehension processes associated with targeted interaction moves (e.g., *mirroring*, *stimulation*, *thinking aloud*). Results indicate strong alignment with the pedagogical strategies. Specifically, 69.0% of responses were rated 4–5 for stimulating reasoning beyond literal information (with 34.5% at the maximum rating), and 72.4% were rated 4–5 for supporting inference formulation (with 44.8% at the maximum rating). Overall, these findings suggest that the automated interactions effectively foster inferential and causal reasoning, in line with the intended instructional goals.

Robustness to instructional drift. We evaluated interaction robustness by examining whether CLAIRe adhered to the predefined pedagogical schema and avoided off-task or hallucinated content. To this end, we used 150 *regular* and 150 *jailbreak* prompts from the *In-The-Wild Jailbreak Prompts* dataset [29], injected at random positions in simulated tutoring dialogues. CLAIRe maintained instructional alignment in all cases, although it occasionally failed to re-anchor the interaction to reading comprehension after rejecting off-task requests (9.3% for *jailbreak* prompts vs. 2% for *regular* prompts). Overall, these results indicate that CLAIRe is generally robust to instructional drift, with minor limitations in explicit re-anchoring.

5.3 Discussion and Future Work

The performed evaluation has highlighted several strengths of CLAIRe. Teachers generally perceive the system as delivering high-quality tutoring, with clear and supportive communication, and questions that foster attentive and reflective reading. Task-oriented assessment confirms that CLAIRe faithfully enacts the teacher-defined guided reading methodology, generating interactions aligned with higher-level comprehension strategies and maintaining pedagogical control even under adversarial prompts. Importantly, observed adaptivity arises from the LLM and prompting strategy rather than the framework’s rule-based logic, showing that teacher-authored procedures can remain stable while adaptive behavior evolves through improvements in models or prompts.

At the same time, the evaluation revealed limitations that motivate future work. Interaction coherence remains a key challenge: lower ratings for response relevance suggest that the current chat-history management, providing the entire conversation to the LLM, can introduce verbosity and irrelevant context, reducing alignment with students’ immediate contributions. More selective history handling or advanced prompting strategies could improve sustained coherence. Additionally, exploring alternative LLMs (e.g., GPT-5 variants) and refined prompt tuning may further enhance evaluation fidelity, adaptivity, and response coherence, while balancing latency and computational costs. Finally, while the current evaluation focused primarily on teacher perceptions, aiming first to validate the system as a reliable and pedagogically faithful tool rather than to assess direct learning outcomes, future classroom studies will be needed to complement perception-based measures and quantify instructional effectiveness.

More broadly, the results highlight how CLAIRE can take over the operational enactment of teacher-defined instructional procedures, allowing the pedagogical work carried out by teachers at design time to be applied consistently and at scale across interactions and learners.

6 Conclusion

We introduced CLAIRE, a teacher-configurable framework that operationalizes guided reading by explicitly representing phases, interaction moves, assessment criteria, and transition logic, executed at runtime through LLM-powered agents. Evaluation shows that the system reliably enacts instructional strategies, supports higher-level comprehension, and is perceived by teachers as pedagogically meaningful and useful. Future work will focus on improving interaction coherence, exploring more advanced language models, and assessing the impact on student learning outcomes in classroom settings.

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